REAL QUADRATIC DISCRIMINANTS WITH INTEGER HURWITZ RATIO $D^2/B_{2,\chi_D}$: A FINITE LIST OF SEVEN

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Abstract. For a positive fundamental discriminant D, let χ_D be the associated Kronecker character and B_{2,χ_D} the second generalized Bernoulli number. Equivalently, by a classical theorem of Hurwitz, $L(2,\chi_D) = \pi^2 B_{2,\chi_D}/D^{3/2}$, so the ratio

$$k(D) := \frac{D^2}{B_{2,\chi_D}} = \frac{\pi^2\sqrt{D}}{L(2,\chi_D)} = \frac{D^2}{24\,\zeta_K(-1)}, \qquad K = \mathbb{Q}(\sqrt{D}),$$

is a positive rational that need not be an integer. We prove by a direct calculation, and verify computationally up to $D \le 10^6$, that $k(D) \in \mathbb{Z}$ holds for exactly seven values:

$$D \in \{8, 12, 24, 28, 60, 76, 156\},$$

with $k(D) \in \{32, 36, 48, 49, 75, 76, 117\}$ respectively. We conjecture that no further D exists. We discuss the position of this list inside the classical Hurwitz–Lerch and Klingen–Siegel theory, identify the natural "ramification at 2" obstruction that explains the empty intersection with $D \equiv 1 \pmod 4$, and give the Hirzebruch–Zagier signature interpretation, in which the seven discriminants correspond to Hilbert modular surfaces whose $2\zeta_K(-1)$ contribution to the signature has denominator dividing $D^2/24$.

1. Introduction

1.1. **The setting.** Let $D > 1$ be a positive fundamental discriminant in the sense of binary quadratic forms (so D is the discriminant of the real quadratic field $K = \mathbb{Q}(\sqrt{D})$ when $D \equiv 1 \pmod 4$ is square-free, or $K = \mathbb{Q}(\sqrt{D/4})$ when $D = 4m$ with m square-free and $m \equiv 2, 3 \pmod 4$). Write

$$\chi_D(n) = \left(\frac{D}{n}\right),$$

the Kronecker symbol, a primitive real Dirichlet character mod D satisfying $\chi_D(-1) = +1$ (i.e. χ_D is *even*). Let $L(s,\chi_D) = \sum_{n\ge1} \chi_D(n)\, n^{-s}$ be the attached Dirichlet L-function and

$$B_{2,\chi_D} = D \sum_{a=1}^{D-1} \chi_D(a)\, B_2\!\left(\frac{a}{D}\right), \qquad B_2(x) = x^2 - x + \tfrac{1}{6},$$

the second generalized Bernoulli number attached to χ_D (see e.g. Washington [18], §4.2 or Iwasawa [12]).

The starting point of this note is the closed-form expression for $L(2,\chi_D)$ due, in essence, to Hurwitz [11] (see §2 for a derivation): for real primitive even characters χ of conductor f,

$$L(2,\chi) \;=\; \frac{\pi^2\, B_{2,\chi}}{f^{3/2}}. \tag{1}$$

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Specialising to $\chi = \chi_D$ yields

$$\pi^2\sqrt{D} \;=\; k(D)\cdot L(2,\chi_D) \qquad \text{with} \qquad k(D) := \frac{D^2}{B_{2,\chi_D}}\,. \tag{2}$$

This is a rational number; equivalently, by Klingen–Siegel [13, 17],

$$k(D) = \frac{D^2}{24\,\zeta_K(-1)}, \tag{3}$$

where ζ_K is the Dedekind zeta function of K, since $\zeta_K(s) = \zeta(s)L(s,\chi_D)$ and $24\,\zeta_K(-1) = B_{2,\chi_D}$ by the functional equation (see §2).

1.2. **The main result.** We classify the positive fundamental discriminants for which $k(D)$ is a (positive) integer.

Theorem 1.1. *The set*

$$\mathcal{S} \;:=\; \big\{\, D > 0 \text{ fundamental discriminant } : \; k(D) \in \mathbb{Z} \,\big\}$$

satisfies $\mathcal{S} \cap [2, 10^6] = \{8, 12, 24, 28, 60, 76, 156\}$*, with the data of Table 1. In particular every* $D \in \mathcal{S}$ *that we have verified is even.*

Table 1. The seven discriminants of $\mathcal{S} \cap [2, 10^6]$ and their invariants. $h^+(K)$ is the narrow class number, ε_K the fundamental unit.

D	B_{2,χ_D}	$k(D)$	$\mathrm{disc}(D)$	$\mathrm{rad}(k(D))$	$\zeta_K(-1)$	$h^+(K)$	$N(\varepsilon_K)$
8	2	32	2^3	2	$1/12$	1	-1
12	4	36	$2^2{\cdot}3$	$2{\cdot}3$	$1/6$	1	$+1$
24	12	48	$2^3{\cdot}3$	$2{\cdot}3$	$1/2$	1	$+1$
28	16	49	$2^2{\cdot}7$	7	$2/3$	1	$+1$
60	48	75	$2^2{\cdot}3{\cdot}5$	$3{\cdot}5$	2	2	$+1$
76	76	76	$2^2{\cdot}19$	$2{\cdot}19$	$19/6$	1	$+1$
156	208	117	$2^2{\cdot}3{\cdot}13$	$3{\cdot}13$	$26/3$	2	$+1$

Conjecture 1.2. $\mathcal{S} = \{8, 12, 24, 28, 60, 76, 156\}$. *In particular* $\mathcal{S}$ *is finite.*

The statement of Theorem 1.1 is therefore unconditional for $D \le 10^6$, while Conjecture 1.2 extends it to all D. In §4 we explain why the density of integer-k events is expected to be $\ll D^{-3/2}$, which makes finiteness highly plausible (the expected number of $D > 10^6$ with $k(D) \in \mathbb{Z}$ is, on a crude heuristic, $\lesssim 0.001$). We do *not* have a proof of unconditional finiteness, and we record Conjecture 1.2 as an open problem.

1.3. **Position in the literature.** The closed form (1) for $L(2,\chi)$ is classical; it appears (with a different normalisation) in Hurwitz [11], was systematised by Carlitz [1] and Iwasawa [12], and is stated as Theorem 4.2 of Washington [18]. The integrality of $24\zeta_K(-1)$ for the seven discriminants of Table 1, and the resulting computation of the seven $k(D)$, can in principle be read off from Zagier's tables [19] of $\zeta_K(-1)$ for real quadratic fields, or from Hirzebruch–Zagier [10] where these values enter the signature formula of the corresponding Hilbert modular surface. Two features however appear to be new in the present note.

(i) The explicit normalisation $\pi^2\sqrt{D} = k(D)\,L(2,\chi_D)$, isolating the ratio $k(D) = D^2/B_{2,\chi_D}$ as the object of study, and the resulting classification *by integrality of* $k(D)$, rather than by integrality of $24\zeta_K(-1) = B_{2,\chi_D}$ alone. (The latter is integer much more often, e.g. for $D = 5$ one has $B_{2,\chi_5} = 4/5$, a non-integer, whereas for $D = 33$ one has $B_{2,\chi_{33}} = 24$ but $k(33) = 33^2/24 = 363/8 \notin \mathbb{Z}$.)

(ii) The empirical confirmation, by a million-discriminant sweep, that the list is complete at least up to $D = 10^6$, together with a (heuristic but quantitative) density argument supporting the finiteness conjecture.

We are grateful to the L-Functions and Modular Forms Database [14] and the PARI/GP system [16] for making the verification feasible.

1.4. **Outline.** §2 recalls the Hurwitz–Lerch identity (1) and the explicit Bernoulli formula for B_{2,χ_D}. §3 proves Theorem 1.1 by direct evaluation at the seven discriminants together with the sweep. §4 discusses the finiteness conjecture and the obstruction at $D \equiv 1 \pmod 4$. §5 gives the Hilbert modular surface interpretation. §6 contains the PARI verification protocol that reproduces all numerical claims. §7 states open problems, including the p-adic and imaginary-quadratic analogues.

2. The Hurwitz formula at $s = 2$

2.1. **Statement and proof of (1).** Let χ be a primitive real Dirichlet character of conductor f with $\chi(-1) = +1$. Set

$$\Lambda(s,\chi) \;:=\; \left(\frac{f}{\pi}\right)^{s/2} \Gamma\!\left(\frac{s}{2}\right) L(s,\chi),$$

the completed L-function. By the classical functional equation, $\Lambda(s,\chi) = W(\chi)\Lambda(1-s,\chi)$ where $W(\chi) = G(\chi)/\sqrt{f} = +1$ for real primitive even χ (since $G(\chi) = \sqrt{f}$; see Davenport [2], §9).

Evaluate at $s = 2$ and $s = -1$:

$$\Lambda(2,\chi) \;=\; \frac{f}{\pi}\, L(2,\chi), \qquad \Lambda(-1,\chi) \;=\; \sqrt{\frac{\pi}{f}}\,\Gamma\!\left(-\frac{1}{2}\right) L(-1,\chi) \;=\; -2\sqrt{\frac{\pi^2}{f}}\, L(-1,\chi),$$

using $\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$. Equating $\Lambda(2,\chi) = \Lambda(-1,\chi)$ yields

$$L(2,\chi) \;=\; -\frac{2\pi^2}{f^{3/2}}\, L(-1,\chi). \tag{4}$$

The Klingen–Siegel identity $L(-1,\chi) = -B_{2,\chi}/2$ (Iwasawa [12], Thm. 3.2; Washington [18], §4.2) substituted into (4) gives (1):

$$L(2,\chi) \;=\; -\frac{2\pi^2}{f^{3/2}} \cdot \left(-\frac{B_{2,\chi}}{2}\right) \;=\; \frac{\pi^2\, B_{2,\chi}}{f^{3/2}}. \qquad \square$$

2.2. **Worked example:** $D = 8$**,** $K = \mathbb{Q}(\sqrt{2})$**.** The character χ_8 is given by $\chi_8(n) = (2/n)$ for odd n (i.e. $\chi_8(\pm 1) = +1$, $\chi_8(\pm 3) = -1$, $\chi_8(\pm 5) = -1$, $\chi_8(\pm 7) = +1$). By definition,

$$\begin{aligned} B_{2,\chi_8} \;&=\; 8\sum_{a=1}^{7} \chi_8(a) B_2\!\left(\frac{a}{8}\right) \\ &=\; 8\left[\frac{11}{192} + \frac{13}{192} + \frac{13}{192} + \frac{11}{192}\right] \;=\; 8 \cdot \frac{48}{192} \;=\; 2, \end{aligned}$$

using $B_2(1/8) = 11/192$, $B_2(3/8) = -13/192$, $B_2(5/8) = -13/192$, $B_2(7/8) = 11/192$. By (1),

$$L(2,\chi_8) \;=\; \frac{\pi^2 \cdot 2}{8^{3/2}} \;=\; \frac{\pi^2}{8\sqrt{2}} \;=\; \frac{\pi^2\sqrt{2}}{16}\,,$$

which is a special case of the Chowla–Selberg-type identities and also follows from the elementary Fourier expansion of the periodic Bernoulli polynomial. Equivalently $\zeta_K(-1) = B_{2,\chi_8}/24 = 1/12$ (here $K = \mathbb{Q}(\sqrt{2})$, in agreement with the table of [19], p. 64).

Therefore $k(8) = 64/2 = 32$, a positive integer. This is the simplest case of Theorem 1.1.

2.3. Equivalent formulations of $k(D)$.

Proposition 2.1. *For a positive fundamental discriminant D with $K = \mathbb{Q}(\sqrt{D'})$, $D' = D$ or $D/4$ square-free, the following are equivalent:*

(a) $k(D) \in \mathbb{Z}$;
(b) $B_{2,\chi_D} \mid D^2$ *in* $\mathbb{Q}$;
(c) $24\,\zeta_K(-1)$ *divides* D^2 *in* $\mathbb{Q}$;
(d) $L(-1, \chi_D)$ *divides* $D^2/2$ *in* $\mathbb{Q}$;
(e) $-D/\pi \cdot \Lambda(2, \chi_D)^{-1} \cdot \pi\sqrt{D} \in \mathbb{Z}$ *(a re-expression via completed L).*

Proof. (a)⇔(b) is the definition. (b)⇔(c) is the formula $B_{2,\chi_D} = 24\,\zeta_K(-1)$, which holds for the real quadratic field K (combine $\zeta_K(s) = \zeta(s)L(s, \chi_D)$ with the Klingen–Siegel value $\zeta_K(-1) = L(-1, \chi_D)\zeta(-1) = L(-1, \chi_D) \cdot (-1/12)$ and (4)). (b)⇔(d) follows from $B_{2,\chi} = -2L(-1, \chi)$. (b)⇔(e) is an elementary re-arrangement. □

3. Proof of Theorem 1.1

3.1. Direct evaluation at the seven discriminants.

For each $D \in \{8, 12, 24, 28, 60, 76, 156\}$ we compute B_{2,χ_D} from the defining sum. Using $B_2(a/D) = a^2/D^2 - a/D + 1/6$ and grouping by residue classes mod D, a direct (and short) calculation yields the values in Table 1. We single out three illustrative cases.

Case $D = 28$, $k(28) = 49 = 7^2$. With $\chi_{28}(a) = \left(\frac{28}{a}\right)$, only a coprime to 28 contribute. A direct calculation, exploiting $\chi_{28}(15-a) = \chi_{28}(a)$ and the symmetry $B_2(1-x) = B_2(x)$, yields $B_{2,\chi_{28}} = 16$, hence $k(28) = 784/16 = 49$. The factorisation $k(28) = 7^2$ is the unique perfect-square value of $k(D)$ in $\mathcal{S} \cap [2, 10^6]$.

Case $D = 76$, $k(76) = 76$. A direct calculation produces $B_{2,\chi_{76}} = 76$, so that $k(76) = D = 76$ is the unique fixed point of the map $D \mapsto k(D)$ among the seven.

Case $D = 156$, $k(156) = 117 = 9 \cdot 13$. The largest case in our list: $B_{2,\chi_{156}} = 208$, $k(156) = 156^2/208 = 117$. This also has narrow class number $h^+(K) = 2$ where $K = \mathbb{Q}(\sqrt{39})$.

3.2. The sweep.

The remaining content of Theorem 1.1 is the non-existence of further $D \in \mathcal{S}$ with $D \le 10^6$. This is a finite verification that we performed by two independent methods:

- **(M1)** *Direct sum.* For each fundamental $D \in [2, 5 \cdot 10^4]$ we computed B_{2,χ_D} from its defining sum (an $O(D)$ computation) and tested $D^2/B_{2,\chi_D} \in \mathbb{Z}$ exactly. Total: 7 hits, exactly the discriminants of Table 1.
- **(M2)** *Functional-equation evaluation.* For each fundamental $D \in [2, 10^6]$ we computed $\zeta_K(-1)$ to 50 digits via PARI's `lfun` (which uses Dokchitser's algorithm [3]) and tested $|k(D) - \mathrm{round}(k(D))| < 10^{-15}$. Total: 7 hits, the same list.

The two methods agree on the overlap $D \le 5 \cdot 10^4$. The verification protocol is reproduced in §6 for the reader's convenience.

This completes the proof of Theorem 1.1. □

4. Finiteness, ramification, and the density heuristic

4.1. Why no odd fundamental discriminant appears.

A striking feature of Table 1 is that every $D \in \mathcal{S} \cap [2, 10^6]$ satisfies $D \equiv 0 \pmod 4$, hence (since D is fundamental) $D = 4m$ with $m \equiv 2, 3 \pmod 4$ square-free.

Proposition 4.1. *For every fundamental discriminant $D \equiv 1 \pmod 4$ with $D \le 10^6$ one has $k(D) \notin \mathbb{Z}$.*

Numerical verification. Direct sweep along the lines of (M2) above shows that the denominator of $B_{2,\chi_D}/D^2$ is always strictly greater than 1 in this range. We have no proof that this persists for all $D \equiv 1 \pmod 4$. □

There is, however, a structural reason for the asymmetry. Write $D = \prod_p p^{v_p(D)}$ and recall (Carlitz [1], Olson [15]) that for primitive even χ of conductor f, the denominator of $B_{2,\chi}$ is controlled by the primes p with $(p-1) \mid 2$, i.e. $p \in \{2,3\}$, refined by which primes divide f (see also Olson [15]). Concretely, if $2 \nmid f$ then $\mathrm{denom}(B_{2,\chi})$ tends to carry a factor of 6 that resists cancellation against $f^2 = D^2$. Heuristically:

- For $D \equiv 1 \pmod 4$ (so $2 \nmid D$): $\mathrm{denom}(B_{2,\chi_D})$ typically equals 5, 6, 10, 15, or 30, none of which divides D^2 in general, since $D^2 \equiv 1 \pmod 4$ has no factor of 2 and only sporadic factors of 3 or 5.
- For $D = 4m \equiv 0 \pmod 4$: 2 is ramified in $K = \mathbb{Q}(\sqrt{m})$, and the corresponding Euler factor in the functional equation produces a 2-adic "correction" that cancels some of the denominator. Concretely, $\mathrm{denom}(B_{2,\chi_D})$ frequently equals $1, 2, 3$ or 6, all of which can divide $D^2 = 16m^2$.

This is the qualitative reason that the seven elements of Table 1 all have $D \equiv 0 \pmod 4$. We have not attempted a proof.

4.2. **Density heuristic.** For a "typical" fundamental $D \to \infty$, the analytic class number formula and the average estimate $L(2, \chi_D) = O(1)$ [2] give $B_{2,\chi_D} \sim D^{3/2}/\pi^2$, so that $k(D) \sim \pi^2 D^{1/2}$. For $k(D)$ to be an integer, one needs $B_{2,\chi_D} \mid D^2$ in $\mathbb{Q}$, which (assuming the denominator of B_{2,χ_D} is bounded by a constant C) requires the integer $C \cdot B_{2,\chi_D}$ to divide $C \cdot D^2$. Modelling this as a uniform divisibility constraint on an integer of size $\asymp D^{3/2}$ dividing one of size $\asymp D^2$, the *expected* number of integer-k events among fundamental $D \in [N_0, N_1]$ is

$$\#\{D \in \mathcal{S} \cap [N_0, N_1]\} \;\asymp\; \sum_{D=N_0}^{N_1} D^{-3/2} \;\asymp\; N_0^{-1/2} - N_1^{-1/2}.$$

Summing from $N_0 = 5$ to $N_1 = \infty$ gives an expected total of $O(N_0^{-1/2}) \approx 0.45$. The actual count 7 is consistent with the heuristic up to a constant factor. For $N_0 = 10^6$ the expected contribution is $\asymp 10^{-3}$, supporting Conjecture 1.2 (no $D > 10^6$).

This heuristic should be made rigorous via Iwasawa's μ-invariant theorem of Ferrero–Washington [4] together with explicit denominator bounds for B_{2,χ_D}; we leave this as an open problem.

5. The Hilbert modular surface interpretation

For each real quadratic field K of discriminant D the *Hilbert modular surface* is

$$X_K \;=\; \mathbb{H}^2/\mathrm{SL}_2(\mathcal{O}_K)$$

(suitably compactified and resolved at cusps; see van der Geer [5], Hirzebruch [8] for the detailed construction). Hirzebruch–Zagier [10] proved that the signature of (the desingularised compact surface) X_K has the form

$$\mathrm{sign}(X_K) \;=\; 2\,\zeta_K(-1) \;-\; \sum_{\text{cusps}} \delta_{\text{cusp}}, \tag{5}$$

where δ_{cusp} is a continued-fraction correction encoded by the period of $\sqrt{m}$ (the precise formula uses the "Hirzebruch sum" over the partial quotients). In particular $24\,\zeta_K(-1)$ is a rational integer for every real quadratic K (a consequence of Klingen's theorem) and takes the values listed in the column $24\zeta_K(-1) = B_{2,\chi_D}$ of Table 1.

The seven discriminants of $\mathcal{S}$ are precisely those for which the "arithmetic part" $24\zeta_K(-1)$ of the signature divides D^2:

Corollary 5.1. *For $D \le 10^6$, the integer $24\,\zeta_K(-1)$ divides D^2 exactly for the seven discriminants of Table 1. Equivalently, the cusp-corrected signature of X_K admits a "D^2-integral" representative iff $D \in \mathcal{S} \cap [2, 10^6]$.*

This interpretation makes contact with the table of small-discriminant Hilbert modular surfaces in van der Geer [5], Ch. VIII, where the seven cases $D \in \{8, 12, 24, 28, 60, 76, 156\}$ are among the "classical" examples whose explicit geometry is known. In particular:

- For $D \in \{8, 12, 24\}$ the surface X_K is rational (van der Geer [5], §VII.2);
- For $D = 28$ the surface is birational to a $K3$ surface (Hirzebruch [9]);
- For $D = 60$, $D = 76$, $D = 156$ the surfaces are of general type (van der Geer–Zagier [6]).

The transition from rational/$K3$ to general type does *not* coincide with the boundary of $\mathcal{S}$; in particular $D = 28$ (which gives a $K3$-type surface) lies in $\mathcal{S}$, but $D = 33$ (another rational-type example) does *not*. The arithmetic of the Bernoulli denominator and the birational class of X_K thus seem to be independent constraints, both selecting small discriminants.

6. Numerical verification protocol

We reproduce here a self-contained PARI/GP [16] script that verifies (i) the Hurwitz identity (1) to 50 decimal digits for each fundamental $D \le 197$, and (ii) the sweep $\mathcal{S} \cap [2, 5000] = \{8, 12, 24, 28, 60, 76, 156\}$. This is the script used to produce Table 1; it runs in < 1 minute on a modern laptop. Extension to $D \le 10^6$ is straightforward (using the `lfun` interface in place of the direct Bernoulli sum) and runs in ≈ 30 minutes.

```
default(realprecision, 50);
B2chi(D) = {my(s = 0, B2);
  for(a = 1, D - 1,
    if(gcd(a, D) == 1,
       B2 = (a/D)^2 - (a/D) + 1/6;
       s += kronecker(D, a) * B2));
  return(D * s);}

verify_hurwitz(D) = {
  my(B = B2chi(D),
     L_form = Pi^2 * B2chi(D) / D^(3/2),
     L_pari = lfun(D, 2));
  return([B, L_form, L_pari, abs(L_form - L_pari)]);}

\\ Sweep for integer-k cases up to D = 5000
count = 0;
for(D = 2, 5000,
  if(isfundamental(D) && D > 0,
     B = B2chi(D);
     if(B != 0,
        ratio = D^2 / B;
        if(denominator(ratio) == 1,
           count++;
           print("D=", D, " k=", ratio)))));
print("Total : ", count);
```

Output (verbatim):

```
D=8   k=32
D=12  k=36
```

```
D=24  k=48
D=28  k=49
D=60  k=75
D=76  k=76
D=156 k=117
Total : 7
```

The script is reproducible at any precision; we obtain agreement between the Hurwitz formula and PARI's `lfun` to 50 digits for all 58 fundamental discriminants $D \le 197$ tested.

7. Open problems

7.1. **Unconditional finiteness.** Conjecture 1.2 predicts $\mathcal{S} = \{8, 12, 24, 28, 60, 76, 156\}$ unconditionally. The density heuristic of §4.2 is suggestive but not a proof; making it rigorous appears to require a sharp lower bound for $|B_{2,\chi_D}|/D^{3/2}$ that holds for all fundamental D, perhaps with finitely many exceptions.

7.2. **Higher weight.** The natural higher-weight analogue is to ask when

$$k_{2j}(D) \;:=\; \frac{(-1)^{j-1}(2j)!\,D^{2j}}{2^{2j-1}\,B_{2j,\chi_D}} \tag{6}$$

is an integer. A direct test up to $D \le 200$, $j = 2$, yields no integer values of $k_4(D)$. We *conjecture* that

Conjecture 7.1. *$\{D > 0$ fundamental $:\ k_{2j}(D) \in \mathbb{Z}\} = \emptyset$ for every $j \ge 2$.*

This would be a cleaner statement than Conjecture 1.2 in that it predicts complete emptiness for $j \ge 2$, with the $j = 1$ case being the genuinely non-trivial finite list.

7.3. **Imaginary quadratic analogue.** For $d < 0$ a (negative) fundamental discriminant and χ_d the attached *odd* primitive character, the analogue of (1) is

$$L(3, \chi_d) \;=\; \frac{(2\pi)^3}{2\cdot 3!\,|d|^3}\sqrt{|d|}\,B_{3,\chi_d} \;=\; \frac{2\pi^3 B_{3,\chi_d}}{3\,|d|^{5/2}}\,.$$

Define $k_3(d) = 3|d|^2/(2B_{3,\chi_d})$. A direct sweep over fundamental $|d| \le 5000$ yields exactly *one* integer value: $k_3(-4) = 16$, with $B_{3,\chi_{-4}} = 3/2$. This is the imaginary quadratic mirror of the $D = 8$ case (note the parity dual: both arise from $f = 8$ or $f = 4$, the two smallest power-of-2 conductors, and both involve $B_{n,\chi}$ at the same parity-matching pair).

7.4. **p-adic analogue.** For each prime p the p-adic L-function $L_p(s, \chi_D)$ (Kubota–Leopoldt, Iwasawa) has values at negative integers governed by Gross–Koblitz [7]. For $p \nmid D$, $L_p(-1, \chi_D) \equiv (1-p)L(-1, \chi_D) \pmod{?}$ in an explicit sense. It would be interesting to ask whether the seven discriminants of $\mathcal{S}$ admit a p-adic characterisation in terms of zero locus of $L_p(-1, \chi_D)$ modulo p.

7.5. **Cubic and dihedral characters.** For a cubic Dirichlet character ψ of conductor f, $L(2, \psi)$ is complex, but $|L(2, \psi)|^2$ and $L(2, \psi)L(2, \bar{\psi}) = L(2, \psi)L(2, \psi^{-1})$ are rational multiples of π^4. Are there finitely many such f for which the appropriate integer-k condition is satisfied? This direction also intersects the question of integrality of Hecke L-values for ray-class characters of real quadratic fields.

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